

SVENETICS : A GWAS pipeline based on SVEN

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The Problem Statement

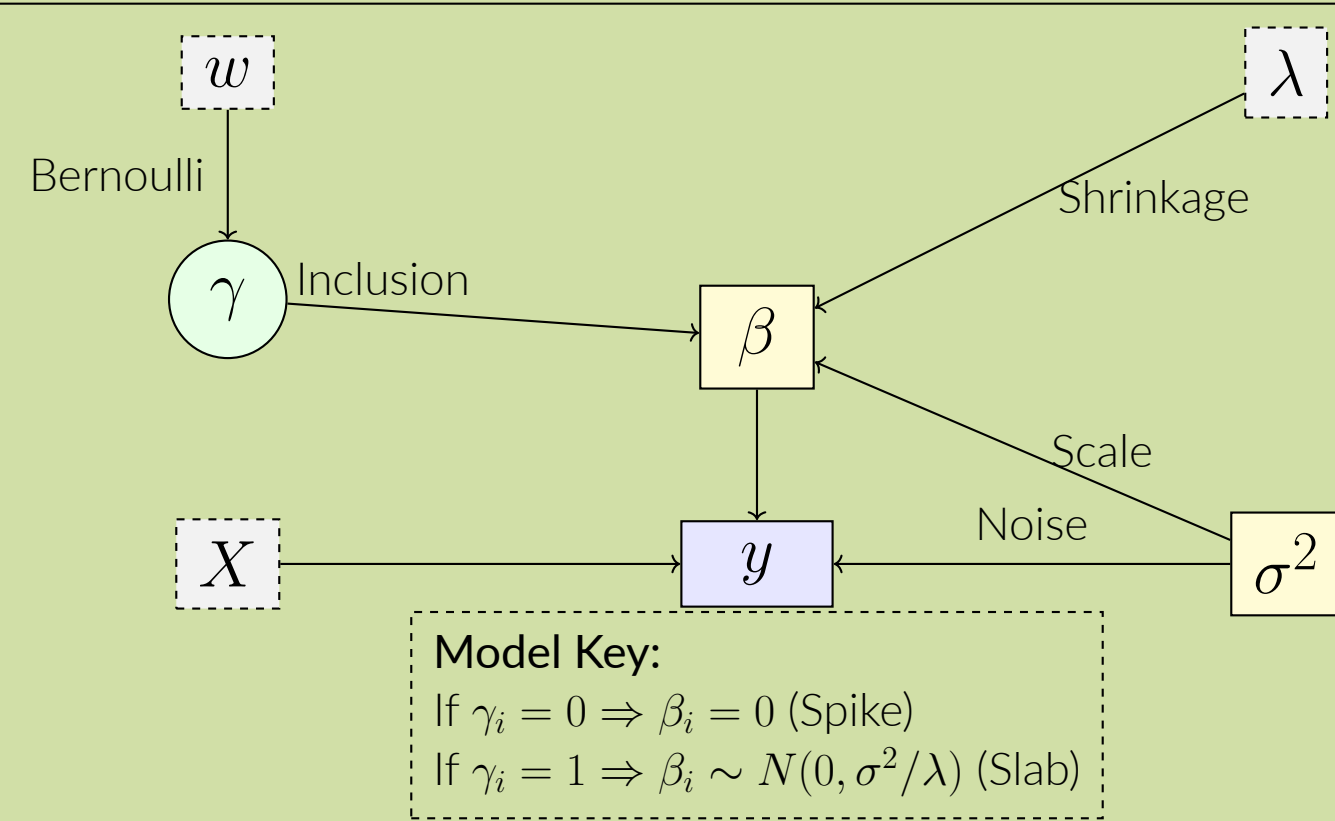
Let us consider that we have n observations on different phenotypic traits of plants. For now, let us focus on one particular trait, say **plant height** and denote the observations by $y = (y_1, y_2, \dots, y_n)$. We also have a matrix X containing p SNPs measured on each of the n plants.

Goal : To study causal links between genetic markers and a phenotypic trait.

Data : $\mathcal{D} = (X, y)$.

Question : Which SNPs play the most significant role in explaining the variability present in the phenotypic trait?

Statistical Model : SVEN



SVEN can be implemented using R package **BRAVO**.

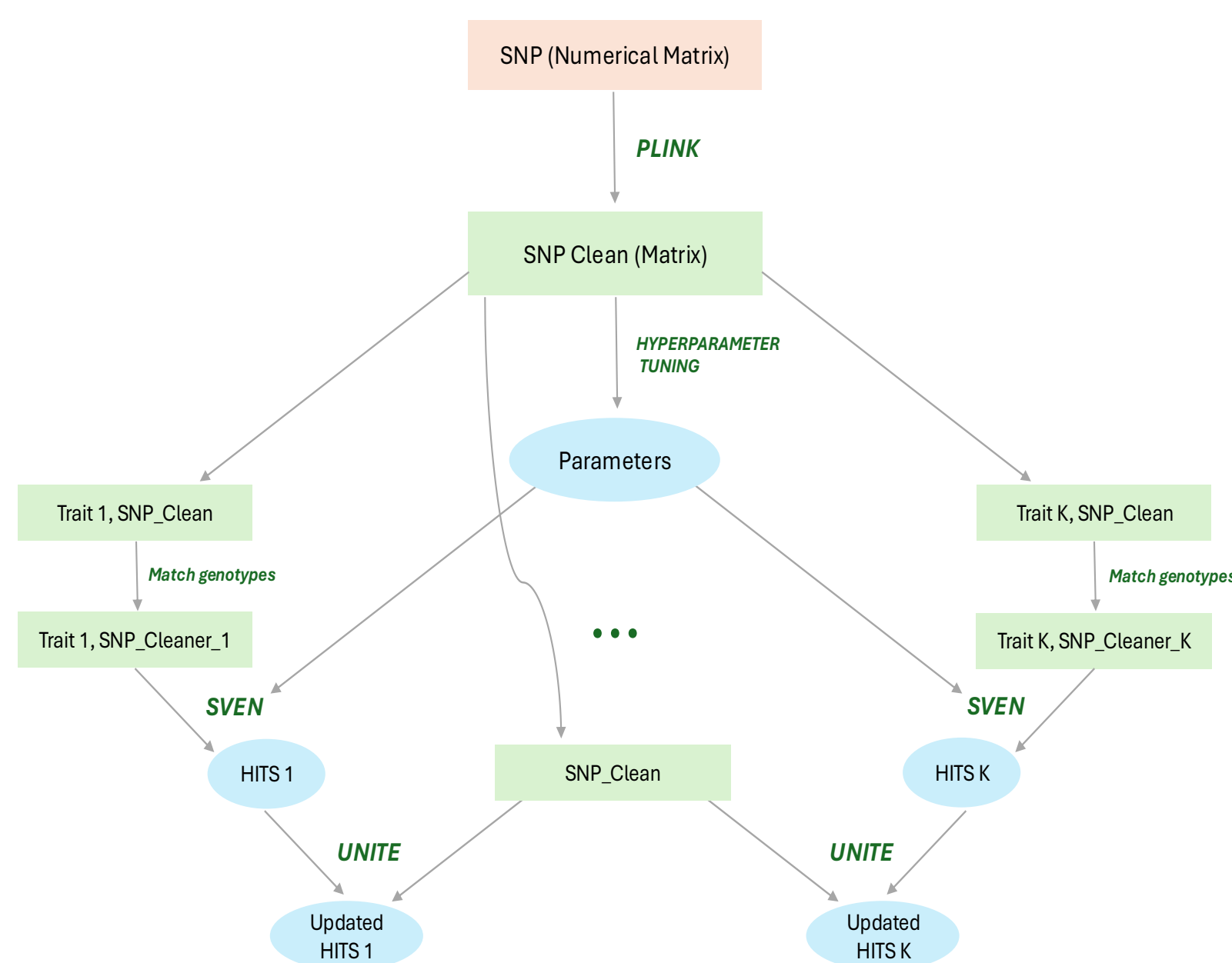
In practice, for the **same genotypic data (SNP matrix)**, we might have **multiple phenotypic traits** that we are interested in.

Tuning parameters in the model : Shrinkage (λ) and Model Size (w)

Objective :

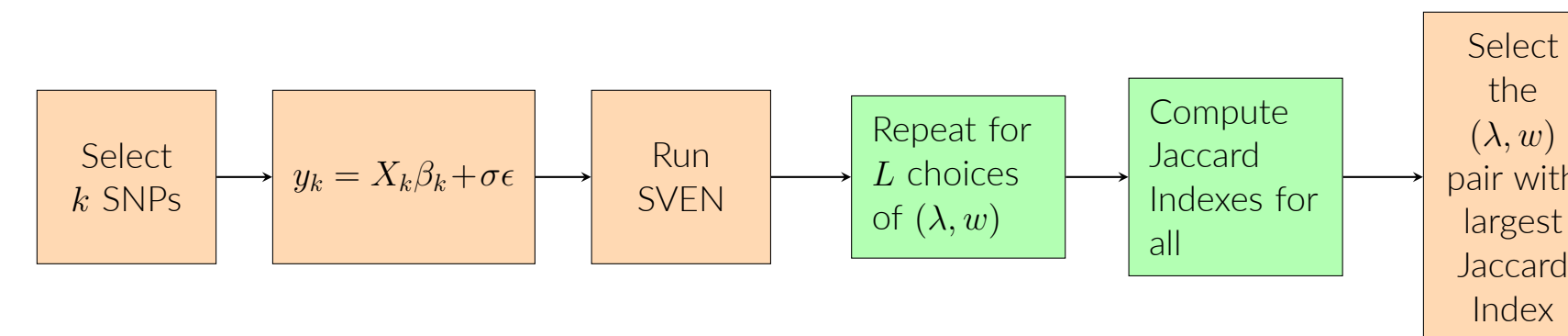
- Choose "Optimal" tuning parameters **only once** which will work for **all traits**.
- Include SNPs which are duplicated / highly correlated among each other.

The GWAS Pipeline : SVENETICS



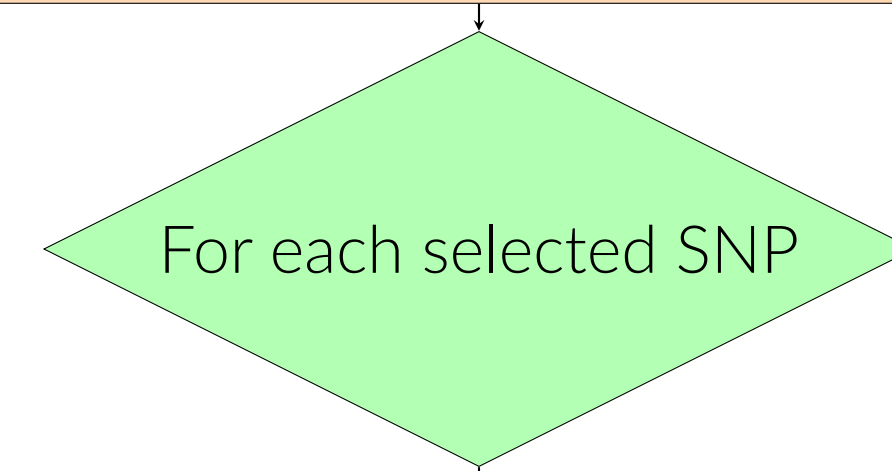
Tuning Hyperparameters

In theory, the tuning parameters λ and w depend only on the design matrix X . So we won't need the response vector y to choose the hyperparameters.



Post processing outputs of SVEN

Step 1: Find SNPs selected by SVEN and their MIPs



Step 2: Find predictors with high correlation ($R^2 > 0.9$)

Step 3: Assign original MIP to correlated predictors

Simulation Experiment

Now we will perform some experiments to compare SVEN with **FARM-CPU** and **BLINK**. For a SNP produced by any method, we call it a **true discovery** if it is highly correlated ($R^2 > 0.9$) with at least one covariate present in the true model, otherwise we call it a **false discovery**. False discovery rate is defined as follows

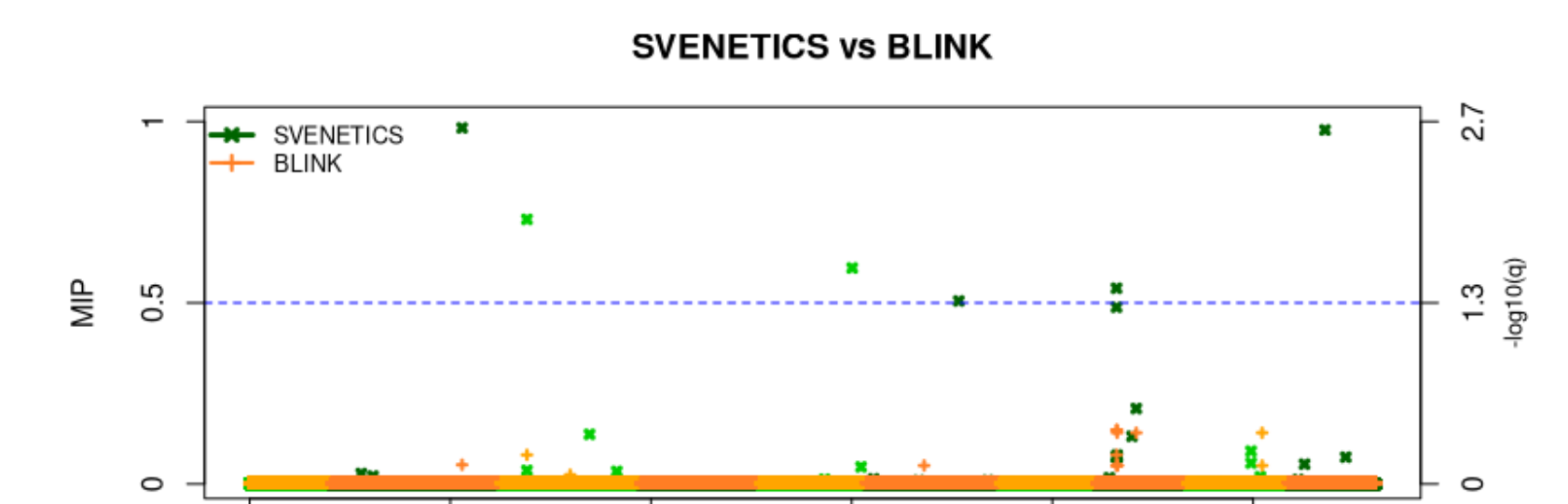
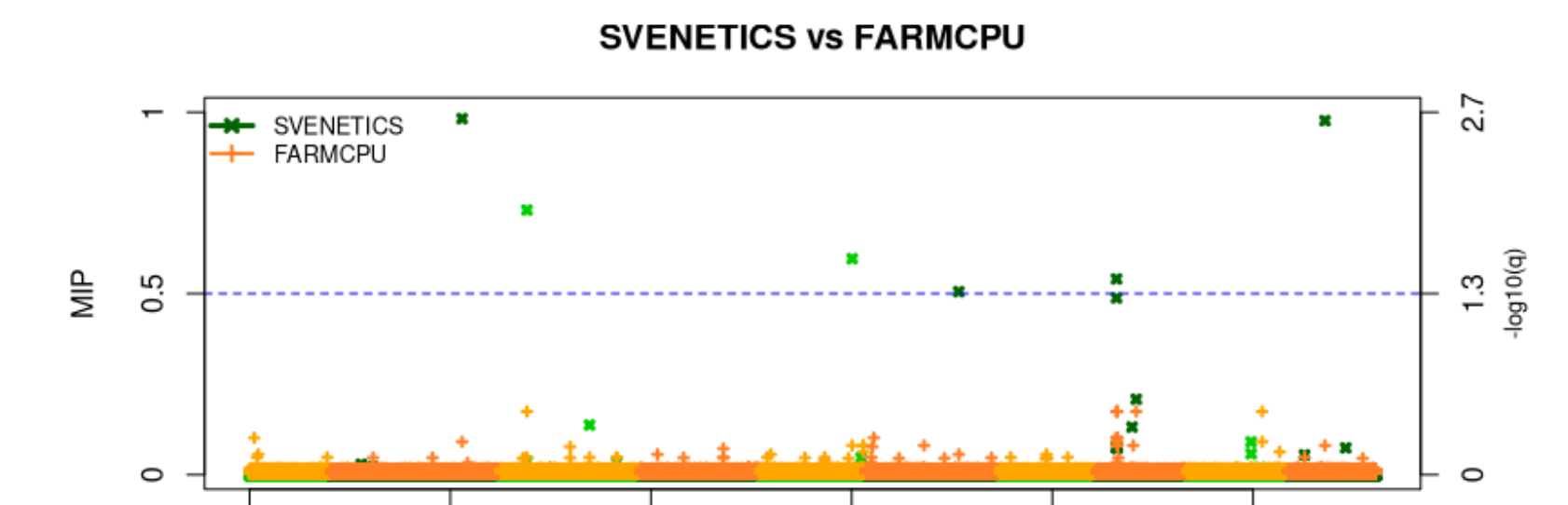
$$\text{FDR} = \frac{\text{no of false discoveries}}{\text{total no of discoveries}}$$

and true positive rate is defined as follows

$$\text{TPR} = \frac{\text{no of true discoveries}}{\text{size of the true model}}$$

SAM volume data analysis

We applied SVENETICS, FARMCPU and BLINK to a data where we have one single trait, namely the Shoot Apical Meristem (SAM) volume.



A Bayesian Measure for SNP importance : P3VE

In genetics, Phenotypic Variance Explained (PVE) by a SNP is defined as percentage of overall variability in Y explained by the that SNP, *without* adjusting for possible effects of other SNPs. Thus, this is simply

$$R^2(Y, X_i) = \frac{\text{Var}(\beta_i X_i)}{\text{Var}(Y)}$$

Let $X_{i*} \in \{0, 1\}$ denote a new value for the SNP and Y_* be the value of the corresponding phenotypic trait. Let $\mathcal{D} = (X, Y)$ denote the data on which the model is trained. **Phenotypic Posterior Predictive Variance Explained** is then defined as follows

$$\text{P3VE}_i = \frac{\text{Var}(\beta_i X_{i*} | \mathcal{D})}{\text{Var}(Y_* | \mathcal{D})} \quad (1)$$

References

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